

Optimal tool setting for a machining centre

by E.A. Pyle

This paper considers the task of an optimal tools setting for a machining centre (MC) to allow a reduction in time spent in the resetting of tools from one batch of workpieces to the next. A theoretical relationship drafted is in full agreement with findings gained as a result of analysis of some 1500 workpieces. A three-dimensional model is presented (in optimal tools setting is offered. A definition of an optimal tools setting; is given, valid for any type of machine-tool.

Keywords: Optimal tools setting, machining centre, machining work pieces

Nomenclature

N_{opt} - number of tools in an optimal tools setting, units
 N_c - number of tools in the core of N_{opt} , units m - number of tools in the replacement part of N_{opt} units
 N_{MS} - number of tool sockets in MS-magazine, N_{MS}^{calc} - calculated surface N_{MS}^{calc} , unit 4×10^{-4} relation of N_{MS}^{calc} to N_{MS} , N_{MS}^{calc} - calculated three-dimensional region N_{MS}^{calc} , unit V_{MS} - theoretical three-dimensional region N_{MS}^{calc} , unit Y_{MS} - immediate surface of projection of N_{MS}^{calc} on the X -plane, unit Y_{MS}^{add} - additional surface of projection of N_{MS}^{calc} on the X -plane, when $N_{MS}^{calc} = 0$, unit 2

Abbreviations

OTS - optimal tools setting for an MS
 TS - tools setting
 OMP - operational manufacturing process, illustrating a sequence of machining of work pieces on different machines with their types indicated
 OMP - operational manufacturing process showing a sequence of machining a workpiece at a machine and indicating a tool, cutting conditions, measurement and auxiliary tools and fixtures
 NC - numerical programmed control device to a type of which (the machines of (the NC-, (W-, I INC-, AC-models, etc. are attributed

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ADS - system of automated design of manufacturing processes on the machines with a computer-aid
 OMP - an optimal operational manufacturing process
 OTS - an optimal tools setting

In books on machine-tools, incorporating numerical programmed control, much has been written concerning the machining centre (MC) in the magazines of which some 30 to 150 tools can be stored (Matalin *et al.* 1974; Modzelevski *et al.*, 1971; Kolka and Kuvshinsky, 1973). The last of those often arouses speculations as to the minimum or maximum number of the tool sockets to be designed or used on an MC. To this end an analysis of production processes on some 15000 work-pieces has been made and a relationship shown in Fig 1 has been established (Ratmirov, 1973). As can be seen from Fig 1, for milling, some (S-A) mill-cutters are

needed, while for drilling about 30 tools are necessary, which makes it necessary to plan a tool magazine for a machine with a capacity of 35-39 units ($N_{MS} = 35-39$). But it is necessary to take into consideration here that Curve 3 does not reach the 100% level on the axis of ordinates by some 4% (which means it is of low validity). In view of this, an attempt to establish theoretical parameters for the required number of tool sockets in the MC tool magazine is offered below.

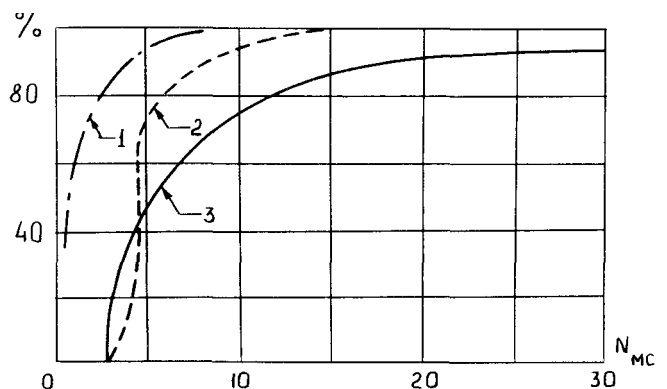


Fig 1 Number of tools for machining a single workpiece: 1 - milling; 2 - turning; 3 - drilling

The designated task is achieved with the help of an optimal tools setting (OTS) for an MC. The following principle has been set down as the basis for the work: there is a certain number of tools stored in the MC tool magazine which are employed in machining all ease-like workpieces on a particular model of the machine with numerical programmed control. Replacement of these tools is carried out only when one or another tool

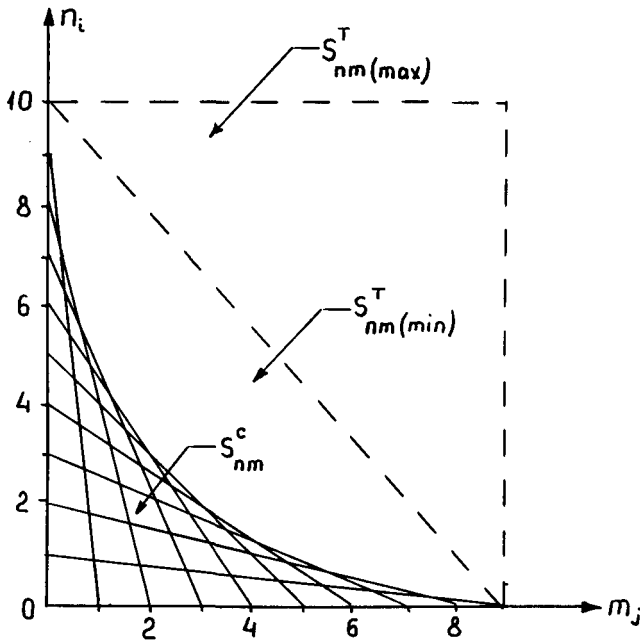


Fig 2 Dependence of the changes in the tool setting core n_i of the increase of the replacement part of the setting m_j

becomes blunted or broken. This part of the tools setting is referred to henceforth as the 'core of the setting'. The ots/c includes also a replacement for a part of the tools setting which may need to be replaced while machining a particular case-like workpiece. Thus, if the core of setting comprises n , tools and the replacement part includes n_{ij} tools, it can be stated that the number of tools in the ots/c should equal

$$N_{ots}/n + m, \quad \dots(1)$$

It should be kept in mind here that $iVots^{MC}$.

The relationship $n_i/(wy)$ is shown in Fig 2. One can see that 10 tools are included in the ots/c that is $N_{ots} = 10$. At this level the values of n , and Wy will have the limits: $n_{\Gamma} = 10$; $n_{\Gamma}'' = 1$; $\Delta^{max} = 9$; $m_{\Gamma}'' = 0$. In other words, the values of n , and m , can be written within the following limits:

$$1/n \cdot SN^{\Delta}; 0 < m^{\Delta} N^{\Delta}, -1.$$

The region S_{nm}^{Δ} shown in Fig 2 represents that region in which changes in the core of setting can take place. The plane 'nom' itself we shall call $1/2, \dots$. To calculate the S_{nm}^{Δ} area the following expression has been derived

$$S^{\Delta} = 0.16(N_{ots}^2 + N_{ots} - 6) \quad \dots(2)$$

An area calculated in accordance with Eqn (2) is equivalent to the calculated surface of $oymc$. If points of $n_{ij} = 1/V_{ots}$, and $m = Ng_{ots} - 1$ (from Fig 2) are connected with a dashed line we obtain a minimum theoretical area $S_{nm}^{\Delta}(min)$ within which a theoretical existence of the $otsuc$ is possible

$$\Delta L(n.in) = 0.5M \gg W_{ots}, -i) \quad \dots(3)$$

Fig 3 illustrates the relationship $qS^{\Delta} = 1/(V_{ots})$, where $qS^{\Delta} = S^{\Delta}/S^{\Delta}$. The theoretically obtained curve from Fig 3 is in a good agreement with the corresponding sum of Curves 1 and 3 shown in Fig 1. If, now, we take the value of qS^{Δ} on the ordinate axis falling between the 2.6- and 3.0-points for 100%

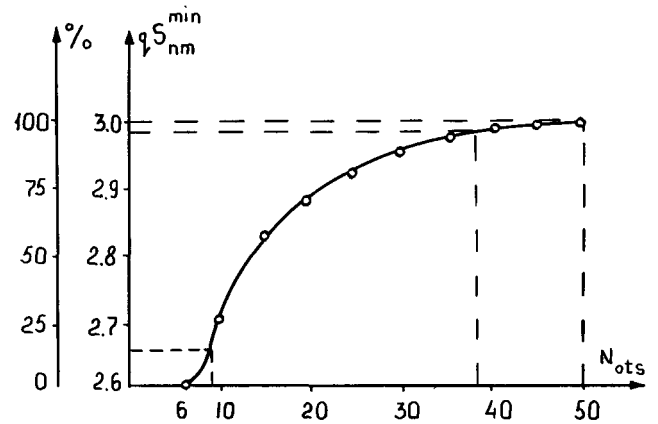


Fig 3 Dependence of changes of the relation of the minimal theoretical region $S_{nm}^T(min)$ to the calculated S_{nm}^c , ie, qS_{nm}^min on a change in N_{ots}

level and plot on the ordinate the above-mentioned figure of 96% and then drop by a perpendicular from this point down to the curve and on to the $1/V_{ots}$ -axis, we then obtain an exact value of the $1/V_{ots} = 38-39$, which is the number looked for. In Fig 3 another axis or ordinates has been drawn for this purpose, characterising the percentage of qS^{Δ} . One can conclude therefore that the theoretical dependence of qS^{Δ} replicates closely a similar curve obtained in practice with a view that the $1/V_{ots}$ -value represents a percentage of work-pieces being machined.

One can derive, on consideration of Fig 3, the practical limits of existence of qS^{Δ} , namely:

$2.6 \leq qS^{\Delta} \leq 3.0$. These limits are obtained for $1/V_{ots} = 6-50$. At the same time, if the number of tools in the MS-magazine is increased to 150 (that is, three-fold) the value of qS^{Δ} changes but by a quantity of 0.09 (or, by 22.5%). One can consider, therefore, that the upper bound for $qS^{\Delta} = 3.0$, and the relationship $qS^{\Delta} = f(1/V_{ots})$ will approach it asymptotically. It should be noted here also that an increase of $1/V_{ots}$ from 6 to 9 leads to a considerable growth in number of the caselike workpieces to be machined but then a gradual decline in the growth of that number is observed.

If a plot similar to that in Fig 3 is constructed but only for the qS^{Δ} , where $qS^{\Delta} = S^{\Delta}/S^{\Delta}$, we obtain a maximum number of tools possible in an MS-magazine. A maximum surface S^{Δ} is obtained through dropping the perpendiculars from points $n = Na, s$ and $m = N_{ots} - 1$. The surface S^{Δ} is calculated according to the formula

$$S^{\Delta} = N^{\Delta}(N_{ots}, -1) \quad \dots(4)$$

The calculations have shown that $1/V_{ots} = 250$. In this case an increase in the number of tools in an MS-magazine up to 1000 pieces (that is, four-fold) gives rise to the $1/V_{ots}$ -value by a figure less than 0.04 (or, by 4%). Thus the upper bound to which the curve qS^{Δ} becomes asymptotic will equal 6.2. The dependence $qS^{\Delta} = f(1/V_{ots})$ is represented in Fig 4.

In view of the fact that the minimum number of tools $1/V_{ots} = 2$ (ie, $n = 1$ and $m = 1$), the theoretical limits for the $2.6 \leq qS^{\Delta} \leq 3.0$ (ie, $1/V_{ots} = 6-50$). The size of changes in the curves of qS^{Δ} from Figs 3 and 4 within the bounds of $1/V_{ots} = 2-5$ has not been considered, since machining of case-like workpieces of even a simple form demands

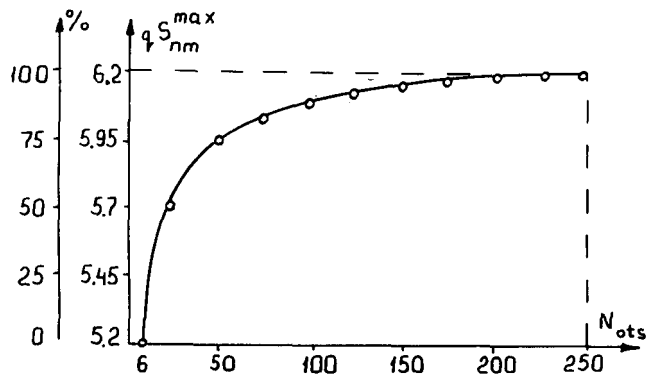


Fig 4 Dependence of changes of the relation of the maximal theoretical region S_{nm}^T to the calculated S_{nm}^c , ie, qS_{nm}^{max} on a change in N_{ots}

the use of several tools, as can also be seen from Fig 1. One can conclude from the above that the ots/c starts to manifest its properties at $11\pi\sigma\delta$.

Fig 5 shows a three-dimensional region of changes in the ots/c , namely $V^\wedge$, where to the x -axis and the w -axis another axis of N_{ots} has been added. As can be seen from Fig 5, the additional planes of $noNy_{..} - 2R_{..}N$ and $moNg_{..} - /?_{..}/v$ have originated. Projections of the $\wedge$ -region on the corresponding planes will characterise the dependence of n , and m on the changes in $Ny_{..}$, that is, $m=f(N^\wedge)$ and $m_1=(M>»)$. Here, the $V^\wedge$ -projection on the $/?_{..}/v$ -plane forms a surface built-up as a sum of the immediate surface $S^\wedge$ and the additional surface $\wedge^\wedge/v$.

$$S^\wedge = S^\wedge + S^\wedge = N^\wedge \quad \dots(5) \text{ where}$$

$$\wedge = \wedge^\wedge (\wedge - 1) \quad \dots(6) \text{ and}$$

$$S_{..}N = N_{..} \quad \dots(7)$$

The projection of $V^\wedge$, on the $R_{..}/y$ is calculated by the formula in the same way as for $S''/\wedge$. The volume

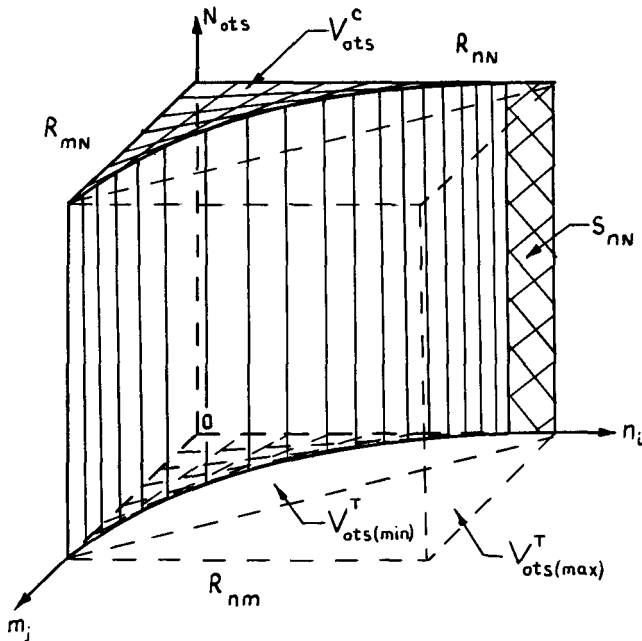


Fig 5 Three-dimensional region ots

occupied by the calculated $V_{..}$, and the theoretical $VJ_{..}$ -region is calculated in the following way

$$V^\wedge = 0, 16N_{..} \{N^\wedge + N^\wedge - 6\} \quad \dots(8)$$

$$V^\wedge(max) = /VL^\wedge ots - 1) \quad \dots(9) \quad \wedge^\wedge(min) = 0.5 \text{ Vott(max)} \quad \dots(10)$$

In the process of increasing of the number of tool sockets in an MS-magazine, the scope of the setting core augments n , while the replacement part of the setting decreases in direct proportion. That is why such a value of N_{Mc}^{can} appear at which the magnitude of the replacement part of ots/c will equal zero ($m = 0$). In other words the ots/c obtained allows the machining of any case-like workpiece in the magazine. This means that switching from one batch of case-like workpieces to a different one will necessitate only a change in the fixtures if that should be necessary. This represents a substantial difference from the group setting of tools (Mitroffanov, 1983). One can conclude from this that the $otSMc^{11}$ represent a set of all group settings of tools (GST), used previously at a particular MS (see Fig 6). It is seen from Fig 6 that the GST represents a set of different tools: $Tu, T, \dots T]$, etc. In some GST, similar tools can be used, as can be seen from Fig 6 (note the dashed branches on the diagram).

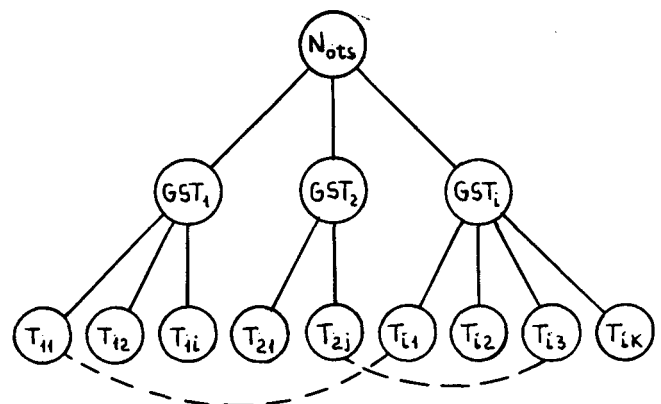


Fig 6 Graph of ots

All the above conclusions are valid for drilling as well as milling machine-tools with numerical programmed control, where $NMC > \wedge$. On the basis of the material described above, a program for a minicomputer has been worked out (Pyl, 1988).

Now consider a problem of the tool-setting for any type of machines, bearing in mind its optimisation. That is, we shall try to give a definition of an optimal tool-setting as such, allowing it to work an article within a minimum time.

It is known that the operational traffic of a manufacturing process (otmp) represents a sequence of transient states of graphs $G_{..} = F(Si)$ - that is, of the operational manufacturing processes starting from a graph of blanking in accordance with a drawing $G^\wedge/_ = F(Sgi_)$ and ending in a graph of an article in accordance with the drawing $Gg_{..} = F(Sy_)$ (Pyl, 1985)

$$G_{..} = F(\wedge) - G = F(\wedge) - \wedge G, =$$

$$F(\wedge) - \wedge \dots G_{..} = F(S_{..}) \dots(11)$$

The schematic representation of Eqn (11) is given in Fig 7.

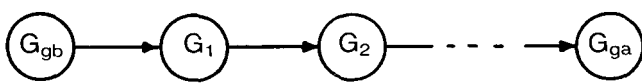


Fig 7 Graph of *otpm*

In other words the *otpm* can be represented as a set of operational manufacturing processes (*ompp*)

$$otmpj = \{omp\}, omp^{\wedge} \dots omp^p \} \quad \dots(12)$$

Here, an *ompp* is a set of different manufacturing operations. Thus, for the MS working on the case-like workpieces the following manufacturing operations are characteristic: drilling operation (DO), countersinking operation (CO), reaming operation (RO) etc.

$$omp, = \{DO, CSO, RO, \dots\} \quad \dots(13)$$

Since to any of such manufacturing operations, a specific tool corresponds, for the drilling operation, therefore, a particular drill (D) will be appropriate (with definite design and technological parameters); to countersinking, a counter-bore (CB), etc. That is, one can write

$$D0, = \{Di, Dz \dots D, \} CSO, = \{CB, CB2 \dots CB, \} \quad \dots(14)$$

We can conclude, therefore, that to each of the *omp*-graphs, $G, \bullet = F(Sj)$, a graph of tool-setting $G, = F'(Sy)$ will correspond, having a tool at the vertex and its design and technical characteristics, as the branches. But here one must take into account that a process of transition of any vertex of the graph *OMP-G*, from a state *S*, to the *S*,+*i*-state can be achieved in several ways with the use of one-, two- or, several tools (see Fig 8).

In view of the above the following definition can be derived:

An operational manufacturing process ompp at the transition of a graph G, of the operational traffic of manufacturing process {otmp,} from a previous state to a subsequent one, G,+|, shall be called optimal when the number of transient vertexes between graphs G, and G,+| is minimal while the number of separately formed and interconnected vertexes of the graph G,+| per unit of

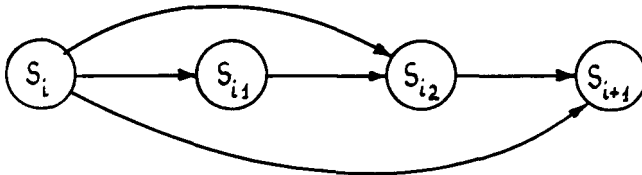


Fig 8 Graph of various ways of machining the ports

time is maximal.

The separate vertexes of the graph *G*,+| represent individual ports the machining of which demands the use of special multi-tool fixtures for a particular case-like workpiece. Such multi-tool fixtures have usually been used previously on the transfer machines but are now found on the new models of MS also.

The interconnected vertexes of the graph *G*,+| imply

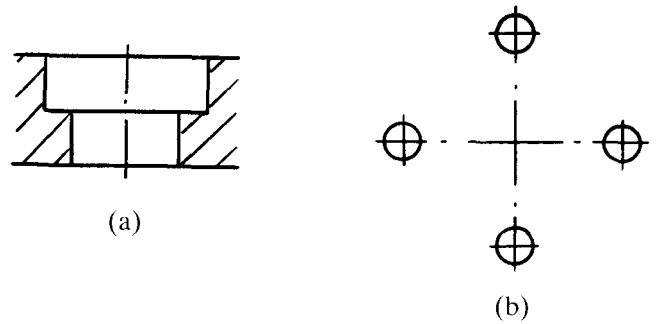


Fig 9 a. Step-like port; b. combination of ports

individual stop-like ports (a sample of one kind of ports is shown in Fig 9a), as well as different combinations of holes (see Fig 9b). For their machining, special tools are employed, together with the multi-tool fixtures adjusted for an appropriate diameter from the CNC-system.

No doubt it would be ideal to unite the machining of separate and interconnected vertexes of the *G*,+|-graph, but if a case-like workpiece is machined on an MS, the flexibility of the latter is lost in working at the combinations of ports.

To each of the optimal operational processes of manufacturing (*ompopi*) an optimal tool setting will correspond (*tSo,,*). But considering the fact that an *ompp* is dependent on particular types of machine-tools employed at a particular factory, it would be more correct to give the following definition:

Tool setting for an operational manufacturing process (ompp) at the transition from a previous state of graph G, of the operational traffic of the manufacturing process (otmp,) to a subsequent state G,+i is to be rendered as optimal if the number of transient vertexes between the graphs G, and G,+| is minimal, while the number of separately formed and interconnected vertexes of the graph G,+| is maximal per unit of time.

Thus, one can say that the tools and machines are primary and the designed *otmp*, and *ompp*, secondary. It follows from this statement that in order to raise the rate of production of workpieces it is necessary to change the stock of machines and/or tools used at a factory, provided, of course, that both the *otmp*, and *ompp* have already been optimal. These changes should, naturally, be a matter of progress - ie, evolutionary.

In consequence, the *omp,,pi* and *dtSgp*, will be optimal only for a particular factory. If a similar workpiece is manufactured at a related factory with a different stock of machines and tools, then for that factory the *omp^p*, and *ts,,p*, will be different. That is, in such cases the existent *otmp*, and *omp?* will not be transferred from one factory to another without an appropriate change. Therefore, this can be written

$$otmp,,,, (omp,,,,) = f(M, T) \quad \dots(15)$$

where: *M* = machines and *T* = tools.

The expression Eqn (15) permits the evaluation of the manufacturing potentials of related factories turning out similar products: that is the more up-to-date the machines and tools employed at a factory, the more efficient the production of workpieces.

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